DeepSeek Math and Prover

Deep Dive into DeepSeek — Chapter 10

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The one rule of this chapter

A stronger checker changes *what* is certified.
It does not certify that the formal object matches the English one.

- An answer scorer, a symbolic checker, and a Lean proof assistant check progressively more structured objects.
- Moving up the ladder strengthens the oracle and raises the specification burden — it never repairs a mismatched formalization.
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Goal: carry one arithmetic equality from a boxed answer to a Lean-checked theorem, and see exactly what each verifier certifies along the way.

Roadmap

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7. Evaluation, failure, and reproducibility

The verification ladder

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Proof acceptance still leaves a separate audit: does the formal statement faithfully represent the English problem?

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Five branches, five different checked objects — none of them interchangeable.

Running example: one equality, four objects

120% **of** 30 **minus** 130% **of** 20

$$\left| \frac{120}{100} \cdot 30 - \frac{130}{100} \cdot 20 \right| = 10.$$

- **English problem:** find the positive difference between the two percentages.

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These four objects are related, but no one of them is identical to another — that is the running thread of the whole chapter.

From concept to code: the formal target

The Prover-V2 README's style of Lean-style theorem for this exact problem has a companion fixture in this book's code release.

from `code/chapter_deepseek_math_and_prover/fixtures/percentage_difference.lean`

```
import Mathlib
import Aesop

theorem percentage_difference :
    abs (((120 : Real) / 100) * 30 -
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  norm_num [abs_of_nonneg]
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`norm_num [abs_of_nonneg]` either leaves zero goals — a checked proof — or it does not. There is no partial credit at the kernel. [INSPECTED: book fixture, `percentage_difference.lean`]

DeepSeek-Math and rule-verifiable rewards

A reward that reads only the boxed answer

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The reward boundary is only the extracted object. A trace that says "130% of 20 is 24" can still end in $\boxed{10}$ and score 1 — correct answer, defective reasoning, both true at once.

DeepSeek-Prover: verified data at scale

Autoformalize, filter, prove, recycle

informal problems

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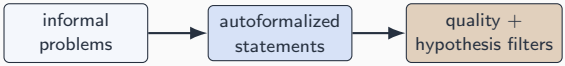

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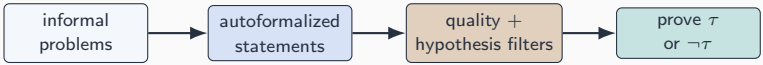

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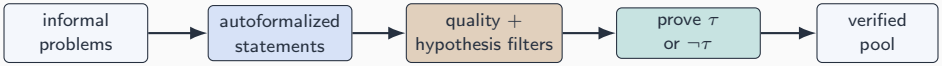

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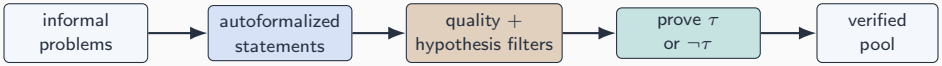

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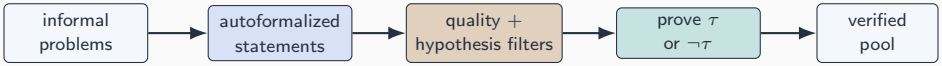

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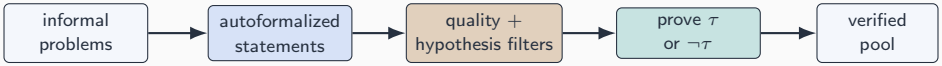

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Acceptance certifies the recorded formal branch, never semantic equivalence with the informal problem.

Prover-V1.5: Lean feedback inside search

From concept to code: checker fields become search state

RMaxTS keeps a tree of proof states and reads structured Lean output to decide what to expand next.

from `DeepSeek-Prover-V1.5/prover/lean/verifier.py`

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result['pass'] = not result['errors']
result['complete'] = (result['pass'] and not result['sorries']
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`pass` and `complete` are different questions: a script can parse (`pass`) while a goal or `sorry` remains (`complete=false`). Only `pass and complete` is a proof success. [INSPECTED: DeepSeek-Prover-V1.5 `prover/lean/verifier.py`]

Prover-V2 and Math-V2: decomposition and self-verification

Two more ways to strengthen the object

Prover-V2: subgoal decomposition

- Rewrites τ into subgoals $\tau_1, \ldots, \tau_m$ plus a composition obligation.
- DeepSeek-V3 decomposes; a 7B prover closes each subgoal.
- Solved subgoals earn credit only when the composition itself checks.

Math-V2: self-verifiable reasoning

- Trains a proof verifier, then uses it as a reward model for the generator.
- Scales verification compute to label hard-to-verify proofs.
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Reported: Prover-V2-671B reaches 88.9% on miniF2F-test and solves 49/658 PutnamBench problems [REPORTED: DeepSeek-Prover-V2 paper].

Evaluation, failure, and reproducibility

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Quoting a benchmark row without its protocol — dataset, split, checker, budget — removes the information needed to compare it to anything else.

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Certify the object you actually checked, and say so on the slide.

Next: the reasoning-RL chapters (R1-Zero, R1) generalize the rule-verifiable reward this chapter specializes to math and proofs.